

BIG-O AND LITTLE-O

The Landau symbolic notation – Big-O and little-o – can be one of the most useful notations in analysis. Learn it well and you become more flexible, quicker, and more sure of yourself in a wide variety of analytic settings.

One concept associated with limits that we weave through this is the notion of “eventually.” To understand this, let $P(x)$ be a statement depending on the variable x that can be either true or false. The statement “as $x \rightarrow 0$, eventually $P(x)$ ” means that $\exists \delta > 0$ such that for $|x| < \delta$, $P(x)$ is true. The statement “as $n \rightarrow \infty$, eventually $P(n)$ ” means that $\exists N$ such that for $n > N$, $P(n)$ is true. Similar statements can be made for all limit situations. As an example: $\lim_{x \rightarrow a} f(x) = L$ iff $\forall \epsilon > 0$, eventually $|f(x) - L| < \epsilon$, where in this case, “eventually” means “ $\exists \delta > 0$ such that for $|x - a| < \delta$.”

With the concept of “eventually” understood, we define of Big-O and little-o. We can define these concepts in any limit situation. In what follows, the function $g(x)$ that we are comparing ought be eventually monotone, nonzero, and should be “reasonably simple”, a concept that we won’t define.

Definition: As $x \rightarrow ?$, $f(x) = o(g(x))$ iff $\lim_{x \rightarrow ?} \frac{f(x)}{g(x)} = 0$.

Definition: As $x \rightarrow ?$, $f(x) = O(g(x))$ iff $\exists C$ such that eventually $|f(x)| \leq C|g(x)|$.

The second definition could be rephrased as saying that the fraction $\frac{f(x)}{g(x)}$ is eventually bounded.

Some examples of this notation:

If $m < n$, as $x \rightarrow \infty$, $x^m = o(x^n)$.

If $m < n$, as $x \rightarrow 0$, $x^n = o(x^m)$.

If $n > 0$, as $x \rightarrow \infty$, $\ln(x) = o(x^n)$.

If $n < 0$, as $x \rightarrow 0^+$, $\ln(x) = o(x^n)$.

As $x \rightarrow \infty$, $\sin x = O(1)$.

As $x \rightarrow 0$, $\sin x = O(x)$.

As $x \rightarrow 0$, $\sin x = x + O(x^3)$.

As $x \rightarrow 0$, $\sin x = x - \frac{x^3}{6} + O(x^5)$.

As $n \rightarrow \infty$, $\sqrt{n^2 + 1} = O(n)$.

As $n \rightarrow \infty$, $\sqrt{n^2 + 1} = n + O\left(\frac{1}{n}\right)$.

This notation is very closely tied up with differentiation and with Taylor polynomials.

For instance, we can prove that the function f is differentiable as x iff $\exists$ a number which we call $f'(x)$ such that as $h \rightarrow 0$, $f(x+h) = f(x) + f'(x)h + o(h)$. That *is* the definition of the derivative right there. The thing we’re calling $o(h)$ is $f(x+h) - f(x) - f'(x)h$ and calling it $o(h)$ is saying that

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x) - f'(x)h}{h} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} - f'(x) = 0.$$

That's what we get is f is differentiable at x . If we have something stronger, in particular that f has a bounded second derivative in a neighborhood of x , then we can say something stronger:

$$f(x+h) = f(x) + f'(x)h + O(h^2).$$

In general, if f has n bounded derivatives in a neighborhood of x , Taylor's Theorem says

$$f(x+h) = \sum_{k=0}^{n-1} \frac{f^{[k]}(x)}{k!} h^k + O(h^n).$$

You'll notice that this last statement is slightly weaker than the full statement of Taylor's Theorem. We have merely said that the error is no worse than some constant times h^n . The full statement of the theorem gives us some tools for estimating how big that constant is; using big-O notation essentially announces that we don't care how big the constant is, as long as it's constant. The reason big-O and little-o is a powerful and flexible notation is that more often than not, we don't care what the constant is.

Manipulating this notation:

Most of what we have here is common sense. Some useful ideas:

$$O(g_1(x)) + O(g_2(x)) = O(\max(g_1(x), g_2(x)))$$

$$g_1(x) \cdot O(g_2(x)) = O(g_1(x) \cdot g_2(x))$$

$$O(g_1(x)) \cdot O(g_2(x)) = O(g_1(x) \cdot g_2(x))$$

We'd like to be able to say that for reasonable functions w , $w(O(g(x))) = O(w(g(x)))$, but we can't always. There's no problem with saying $(O(h))^2 = O(h^2)$, but $e^{O(\ln x)}$ isn't well defined. You have to be careful there.

You should avoid dividing by big-O or little-o. However, one can make sense of something like $\frac{1}{2+O(x)}$ by long division: $\frac{1}{2+O(x)} = \frac{1}{2} + O(x)$ as $x \rightarrow 0$.

Examples:

EXAMPLE 1: Compute $\lim_{x \rightarrow 0} \frac{1 - \cos(x^3)}{\sin^2 x \sin(x^4)}$.

Technically speaking, you could use L'Hôpital's Rule. But I'm not willing to demonstrate that, nor am I willing to help you out if you're foolish enough to try. But this is in fact very quick, using the Taylor series of the functions in question: $\cos u = 1 - \frac{u^2}{2} + O(u^4)$ and $\sin u = u + O(u^3)$. Hence,

$$\begin{aligned} \frac{1 - \cos(x^3)}{\sin^2 x \sin(x^4)} &= \frac{1 - \left(1 - \frac{x^6}{2} + O(x^{12})\right)}{(x + O(x^3))^2 (x^4 + O(x^{12}))} = \frac{\frac{x^6}{2} + O(x^{12})}{(x^2 + O(x^4))(x^4 + O(x^{12}))} \\ &= \frac{\frac{x^6}{2} + O(x^{12})}{x^6 + O(x^8)} = \frac{\frac{1}{2} + O(x^6)}{1 + O(x^2)} \end{aligned}$$

which clearly has the limit $\frac{1}{2}$. To be honest, we spent more effort keeping track of exponents there than we needed to. This is a problem that a reasonably experienced person can work in his or her head.

EXAMPLE 2: If $f(x) = \sqrt{\cos(x)}$, find $f''(0)$ and $f^{[4]}(0)$.

That sounds simple enough, but it can get messy. Let's find the fourth degree Taylor polynomial instead, using the cosine series and the binomial series. The binomial series tells us that

$$(1+u)^\alpha = 1 + \alpha u + \frac{\alpha(\alpha-1)}{2}u^2 + O(u^3)$$

as $u \rightarrow 0$. (Note: if we need more terms than this, we'll find out, and can return to this point to get more terms.)

$$\begin{aligned}\sqrt{\cos x} &= \left(1 - \frac{x^2}{2} + \frac{x^4}{24} + O(x^6)\right)^{\frac{1}{2}} \\ &= 1 + \frac{1}{2} \left(-\frac{x^2}{2} + \frac{x^4}{24} + O(x^6)\right) + \frac{\frac{1}{2}(-\frac{1}{2})}{2} \left(-\frac{x^2}{2} + O(x^4)\right)^2 + O(x^6) \\ &= 1 - \frac{x^2}{4} + \frac{x^4}{48} + O(x^6) - \frac{1}{8} \left(\frac{x^4}{4} + O(x^6)\right) + O(x^6) \\ &= 1 - \frac{x^2}{4} + \left(\frac{1}{48} - \frac{1}{32}\right)x^4 + O(x^6) \\ &= 1 - \frac{x^2}{4} - \frac{x^4}{96} + O(x^6)\end{aligned}$$

We can now read off that $f''(0) = 2! \left(-\frac{1}{4}\right) = -\frac{1}{2}$ and $f^{[4]}(0) = 4! \left(-\frac{1}{96}\right) = -\frac{1}{3}$.

EXAMPLE 3: Find $\lim_{n \rightarrow \infty} \left[\frac{1}{\pi} \int_0^\pi \left(1 + \frac{\cos x}{\sqrt{n}}\right)^{-1} dx \right]^n$

This will provide a good workout for our calculus methods. This is a 1^∞ indeterminate case, so we start by taking the logarithm.

For small u , $(1+u)^{-1} = 1 - u + u^2 + O(u^3)$. (Geometric series).

Thus, $\left(1 + \frac{\cos x}{\sqrt{n}}\right)^{-1} = 1 - \frac{\cos x}{\sqrt{n}} + \frac{\cos^2 x}{n} + O(n^{-3/2})$.

$$\frac{1}{\pi} \int_0^\pi \left(1 + \frac{\cos x}{\sqrt{n}}\right)^{-1} dx = \frac{1}{\pi} \int_0^\pi 1 - \frac{\cos x}{\sqrt{n}} + \frac{\cos^2 x}{n} + O(n^{-3/2}) dx = 1 + 0 + \frac{1}{2n} + O(n^{-3/2}).$$

For small u , $\ln(1+u) = u + O(u^2)$. Thus,

$$\begin{aligned}n \ln \left[\frac{1}{\pi} \int_0^\pi \left(1 + \frac{\cos x}{\sqrt{n}}\right)^{-1} dx \right] &= n \ln \left(1 + \frac{1}{2n} + O(n^{-3/2}) \right) = n \left(\frac{1}{2n} + O(n^{-3/2}) + O(n^{-2}) \right) \\ &= \frac{1}{2} + O(n^{-1/2}). \text{ This goes to } \frac{1}{2}, \text{ so our original limit is } e^{1/2} = \sqrt{e}.\end{aligned}$$

Exercises:

1. Does the series $\sum_{n=1}^{\infty} n^3 [\sin(\arctan(n^{-1})) - \arctan(\sin(n^{-1}))]$ converge or diverge?
2. Suppose that f is defined on an interval, and $\forall y$ in that interval, as $x \rightarrow y$, $f(x) - f(y) = o(|x - y|)$. Prove that f is constant.
3. In all parts of this problem, assume that $f \in C^4$ (that is, has four continuous derivatives).
 - (a) Show that $\frac{f(x+h) - f(x)}{h} = f'(x) + O(h)$ as $h \rightarrow 0$.
 - (b) Show that $\frac{f(x+h) - f(x-h)}{2h} = f'(x) + O(h^2)$ as $h \rightarrow 0$.
 - (c) Show that $\frac{f(x+h) - 2f(x) + f(x-h)}{h^2} = f''(x) + O(h^2)$ as $h \rightarrow 0$
4. True or false: if $a_n = o\left(\frac{1}{n}\right)$ as $n \rightarrow \infty$, then $\sum_{n=1}^{\infty} a_n$ converges. If true, prove. If false, give a counterexample.